

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0517

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

1. Thresholding

(a) Explain Global Thresholding in Segmentation

Global thresholding is one of the simplest image segmentation techniques. It separates an image into **foreground (object)** and **background** by using a **single threshold value (T)** for the entire image.

The process is:

- Choose a threshold value **T**.
- Compare every pixel intensity with **T**.
- If the pixel intensity is **greater than or equal to T**, it is assigned **1 (white/foreground)**.
- If the pixel intensity is **less than T**, it is assigned **0 (black/background)**.

Mathematical Formula:

$$g(x,y)=\begin{cases} 1, & \text{if } f(x,y)\geq T \\ 0, & \text{if } f(x,y)<T \end{cases}$$

where:

$f(x,y)$ = Original pixel intensity

$g(x,y)$ = Output segmented image

T = Threshold value

Advantages

- Easy to implement.
- Fast and computationally efficient.
- Suitable for images with clear contrast between object and background.

Disadvantages

- Does not work well when lighting is uneven.
- Sensitive to noise.
- A single threshold may not segment complex images correctly.

(b) Given Pixel Values

Pixel Value:

[200]

Threshold:

$T = 100$

Rule

- $\text{Pixel} \geq 100 \rightarrow 1$
- $\text{Pixel} < 100 \rightarrow 0$

Calculation

Pixel = 200

Since,

$$200 \geq 100$$

Therefore,

Output = 1

Segmented Output

[1]

Answer: The pixel belongs to the **foreground (object)**.

2. Edge Detection

(a) Explain the Concept of Edge Detection

Edge detection is an image processing technique used to identify the **boundaries of objects** in an image. An edge occurs where there is a sudden change in pixel intensity.

It helps in:

- Object detection
- Image segmentation
- Feature extraction
- Shape recognition
- Medical image analysis
- Computer vision applications

One of the most widely used methods is the **Sobel operator**, which calculates the image gradient in horizontal and vertical directions.

The Sobel operator uses two kernels:

- Horizontal kernel ((G_y))
- Vertical kernel ((G_x))

The gradient magnitude is

$$\sqrt{G} = \sqrt{G_x^2 + G_y^2}$$

A larger gradient value indicates a stronger edge.

Advantages

- Simple and efficient.
- Reduces the effect of image noise.
- Detects strong edges accurately.

Disadvantages

- Less accurate for very noisy images.

- Produces thick edges.
- May miss weak edges.

(b) Apply the Sobel Operator (Horizontal)

Sobel Horizontal Kernel

```
-1 -2 -1
0 0 0
1 2 1
```

Image

```
10 10 10
20 20 20
30 30 30
```

Multiply each element with the corresponding kernel value.

Kernel	Image	Product
-1	10	-10
-2	10	-20
-1	10	-10
0	20	0
0	20	0
0	20	0
1	30	30
2	30	60
1	30	30

Sum of Products

$$(-10)+(-20)+(-10)+0+0+0+30+60+30$$

$$=-40+120$$

$$=80$$

Gradient at the Center

Gradient = 80

Interpretation

The gradient value **80** indicates a noticeable horizontal intensity change, meaning a horizontal edge is present.

3. Region-Based Segmentation

(a) Explain Region Growing in Segmentation

Region growing is a segmentation method that groups neighboring pixels with **similar intensity values** into the same region.

The algorithm starts with one or more **seed pixels**. It checks neighboring pixels and includes them in the region if their intensity difference from the seed is less than or equal to a specified threshold.

Steps

1. Select a seed pixel.
2. Compare neighboring pixels with the seed.
3. Compute the intensity difference.
4. If the difference is within the threshold, include the pixel in the region.
5. Continue until no more pixels satisfy the condition.

Advantages

- Produces connected regions.
- Simple and intuitive.
- Works well when objects have similar intensities.

Disadvantages

- Depends heavily on seed selection.
- Sensitive to noise.
- Can produce incorrect regions if the threshold is not chosen properly.

(b) Given Pixel Intensities

48 50 52

Seed Pixel

50

Threshold Difference

5

Step 1: Compare Each Pixel

For pixel **48**

$$|48-50|=2$$

Since

$$2 \leq 5$$

Include **48**.

For pixel **50**

$$|50-50|=0$$

Since

$$0 \leq 5$$

Include **50**.

For pixel **52**

$$|52-50|=2$$

Since

$$2 \leq 5$$

Include **52**.

Final Answer

All three pixels satisfy the threshold condition, so **48, 50, and 52** belong to the same region.

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterpret s or unable to understand problem	Problem Understanding
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results